

Multifold Fusion Attention Variant for Emotion Recognition

Nikhil Reddy Yerragonda
Department of Computer Science
The University of Alabama at Bhm
 Birmingham, AL, USA
 yreddy00077@gmail.com

Joshua Giddirappa
Department of Computer Science
The University of Alabama at Bhm
 Birmingham, AL, USA
 joshua.giddirappa7@gmail.com

Chetana Krishnan
Department of Biomedical Engineering
The University of Alabama at Bhm
 Birmingham, AL, USA
 krishnac@uab.edu

Omprakash Balaiah Indla
Department of Electrical and Computer Engineering
The University of Alabama at Bhm
 Birmingham, AL, USA
 omprakashindla54@gmail.com

Abstract— Emotion recognition is increasingly critical in applications like artificial intelligence, healthcare, and human-computer interaction, enabling systems to understand and respond to human emotions. Existing methods often rely on pre-extracted features and simplistic fusion strategies, limiting their effectiveness in handling real-world, heterogeneous multimodal data such as audio, video, and Electroencephalography. This paper introduces the multifold fusion attention variant, an end-to-end framework that processes raw data without feature preprocessing, addressing the challenges of existing approaches. Multifold fusion attention variant employs a novel attention-based intermediate fusion mechanism and dynamic modality weighting, enabling it to adapt to varying data quality and capture both short- and long-range dependencies. Tested on the EAV dataset, the multifold fusion attention variant outperforms existing models, achieving an accuracy of 91.2% with a 7.39% improvement over baselines like TransNet. The results demonstrate multifold fusion attention variant robustness and its potential to advance emotion recognition in real-world applications.

Keywords—*Attention, Emotion, Cross-Spatial, Embedding, Multi-Variant, Fusion.*

I. INTRODUCTION

Understanding human emotional states is a critical challenge in machine learning, offering valuable insights into social signals across diverse domains, from AI to human-computer interaction [1], [2]. Researchers have developed numerous approaches and emotion models, addressing tasks such as identifying discrete emotional classes like 'happy,' 'angry,' or 'sad,' as well as estimating emotional dimensions, including excitement and valence, on various continuous scales [3], [4]. This field has explored various methodologies, leveraging a wide array of input modalities such as textual data [5], audio signals [6], and visual information [7].

The rapid growth of available data has spurred the development of a variety of techniques designed to maximize its utility, driving advancements in multimodal methods [8], [9], [10]. Multimodal approaches process multiple data types simultaneously, enabling more comprehensive analysis. Examples include video data combining audio and video streams [11], paired 3-channeled and 3D images [12], and RGB with motion data [13]. Techniques for leveraging such

feature representations range from straightforward result-level fusion to sophisticated feature learning frameworks. While intermediate feature fusion often leads to improved performance by jointly learning representations across modalities, simpler early or late fusion strategies remain widely adopted due to their practicality and flexibility [8]. However, early fusion can struggle with highly heterogeneous data types, and late fusion is limited by treating modality-specific features independently.

Many multimodal methods developed thus far assume the consistent availability of all features during inference and primarily evaluate model performance under such ideal conditions. However, in real/clinical-world scenarios, it is common for modalities to suffer from degraded quality, making robustness to such challenges a critical consideration in designing multimodal systems capable of effectively handling real-time and world data.

With respect to multimodal emotion recognition, particularly with transformer-based frameworks, extracted features are eventually integrated through a learned model and are utilized instead of employing full trainable frameworks [14], [15][16]. While effective in controlled environments, this approach faces limitations in practical applications. Extracting necessary features in unconstrained settings can be complex and introduces additional sources of irrelevancy into the pipeline. This issue is especially pronounced in methods involving language data [14][15], where transcriptions of audio are often unavailable or require separate, potentially error-prone estimation steps.

While other studies in multimodal emotion recognition have predominantly relied on audio-visual modalities, they often overlook the Electroencephalography (EEG) modality, which provides critical insights into intrinsic emotional states through neural activity. This gap exists primarily due to the challenges of integrating EEG data, such as its complex preprocessing requirements and the need for specialized equipment.

Our work addresses these challenges and incorporates EEG as a key modality, setting our approach apart. By leveraging EEG data, we achieve a more comprehensive understanding of emotions, as it captures internal

physiological responses that audio-visual cues alone cannot provide. This inclusion enhances the robustness and accuracy of our system, particularly in scenarios where external factors may distort traditional modalities. As a result, we focus on the problem of audio-video emotion recognition without reliance on separate feature extraction processes, aiming for more practical and robust systems suitable for real-world applications. In this study, we aim to mitigate the limitations of current multimodal emotion recognition methods by developing a full framework that mitigates the need for prior feature extraction, employs intermediate-level fusion, and demonstrates robustness to diverse data scenarios. Our main contributions specifically are:

- **End-to-End Architecture:** We introduce a novel architecture for audiovisual emotion recognition that directly processes raw facial video and speech data without relying on pre-extracted features, enabling seamless end-to-end learning. Additionally, multi-attention is employed to capture long-short dependencies.
- **Innovative Fusion Mechanism:** We implement and evaluate several modality fusion strategies, including a novel attention-based intermediate feature fusion mechanism that dynamically focuses on modality-independent features.
- **Dynamic Modality Weighting:** We propose a new feature weighting mechanism that dynamically adjusts the importance of each modality during inference based on data quality and consistency. This approach allows the model to prioritize more reliable modalities, improving overall performance without requiring explicit dropout mechanisms during training.

Our study acknowledges a few limitations, like fewer sample sizes and inter-observer variability in annotating the dataset, which will be addressed in future work.

II. RELATED WORK

Emotion recognition has become a significant focus in the machine learning community, leading to the development of numerous methods tailored to this task. These approaches utilize various types of data, such as images [7], audio [6], text [5], and human signals [17]. Furthermore, there is a growing trend toward multimodal methods that combine multiple data streams using different fusion strategies [14][16][15]. By leveraging complementary information from various modalities, these techniques aim to improve the accuracy and strength of recognition systems.

In multimodal machine learning, three primary fusion strategies are commonly identified. Early fusion involves combining raw or preprocessed data from different modalities, such as concatenation or addition, and processing them jointly. Late fusion, on the other hand, processes each modality independently, and their outputs, such as feature representations or classification scores, are combined only at the final stage of the network. Intermediate fusion falls between these two, sharing and integrating features at the network's middle layers, allowing for the joint learning of representations across modalities [14][16].

Recent multimodal fusion methods rely on self-attention mechanisms [18], which have proven effective for capturing cross-modal interactions. Self-attention enables the dynamic

modeling of dependencies between modalities, allowing one modality to enhance the representation of another. This concept has been widely applied, particularly in transformer-based architectures, to solve a variety of tasks requiring multimodal fusion.

Several works have utilized transformers for multimodal emotion recognition. For instance, one approach integrates unaligned audio, vision, and text modalities by extracting modality-specific features through convolutional layers and then applying transformer modules for cross-modal fusion [14]. The fused representations are processed further for final classification. Another example focuses on audio-visual emotion recognition, where separate transformers handle each modality before applying cross-modal attention for integration [16]. Unlike earlier methods, this approach fuses one modality into the representation of the other in a single direction. Similarly, another work explores speech and text fusion, using cross-modal attention blocks to learn representations in both directions and subsequently pooling and concatenating them for emotion classification [15].

Despite their effectiveness, these methods exhibit notable limitations. Many rely on pre-built features, such as facial expressions or annotated text, which require separate preprocessing steps [14][15][16]. This dependency on feature extraction increases the complexity of the pipeline and poses challenges in real-world applications, where such preprocessing may not be feasible. Additionally, while these approaches emphasize multimodal fusion, they do not capture short and long-range dependencies. This reliance on pre-extracted features introduces an additional layer of uncertainty and complexity, particularly in unconstrained environments.

To address these limitations, there is a need for approaches that eliminate the dependency on feature preprocessing and enable end-to-end learning directly from raw multimodal data. Such methods would simplify the processing pipeline, enhance the application of emotion recognition systems to real-world and clinical scenarios, and ensure greater robustness across diverse conditions.

III. METHODS

A. Dataset

We employed the publicly available EAV dataset [19]. The EAV dataset is a comprehensive multimodal dataset designed for emotion recognition, featuring synchronized EEG, audio, and video data collected during conversational interactions. It comprises recordings from 42 participants, each contributing 200 interactions balanced across five emotion classes: anger, happiness, sadness, calmness, and neutrality. EEG data was recorded using a 30-channel setup at a sampling rate of 500 Hz, capturing neural activity with high resolution. Audio and video data focused on participants' spoken responses and facial expressions during the scripted conversations. The dataset provides annotations through self-reported valence and arousal scores, supervisor observations, and synchronized class labels, enabling a detailed exploration of emotional states across modalities.

Primary Preprocessing for each modality was performed to ensure usability and consistency. EEG signals underwent filtering to remove noise, down sampling to 100 Hz, and segmentation into 20-second intervals. These were structured into arrays with dimensions $[200 \times 2000 \times 30]$, representing instances, time points, and channels. Audio data was

processed to extract Mel-Frequency Cepstral Coefficients (MFCCs) and other features, which were concatenated into arrays of shape $[100 \times 20 \times 200]$ corresponding to trials, features, and time steps for the speaking tasks. Video data was preprocessed by sampling frames at regular intervals, resulting in sequences of $[200 \times 25 \times H \times W \times C]$, where H , W , and C denote frame dimensions and color channels. These preprocessing steps ensured temporal alignment across modalities and standardized data formats for machine learning applications.

The preprocessed data was converted into pkl (pickle) files to facilitate use in Python-based workflows. Each file contains structured dictionaries with keys corresponding to modalities (EEG, Audio, Video), emotion labels, and values containing the respective data arrays. The EEG data, initially acquired in BrainVision Core Data Format, was transformed into MATLAB (.mat) files before being serialized into pkl. Similarly, audio features and video frame sequences were organized into arrays and stored in .pkl files. This format enables efficient loading, processing, and integration of data into machine learning pipelines, supporting tasks such as multimodal fusion and classification.

The dataset's standardized shapes and formats ensure compatibility with deep learning architectures. For example, EEG data is formatted as $[\text{instances} \times \text{time points} \times \text{channels}]$, audio features as $[\text{instances} \times \text{features} \times \text{time}]$, and video sequences as $[\text{instances} \times \text{frames} \times H \times W \times C]$. These shapes align well with transformer-based models and convolutional neural networks (CNNs), making the EAV dataset highly versatile for emotion recognition tasks. The EAV dataset provides a valuable resource for advancing emotion recognition research in conversational contexts by offering synchronised multimodal data with robust preprocessing and convenient storage formats. The required relevant features were extracted during the training.

Normalization was applied across modalities to standardize the data and enhance the performance of machine learning models. For EEG data, the signals were normalized channel-wise to account for variations in amplitude between participants. This involved rescaling each channel's data to a zero-mean, unit-variance format, effectively reducing biases introduced by inter-subject differences and noise. Audio data was scaled to fall within a consistent range, between 0 and 1 or with mean centering and variance normalization. This ensured that variations in audio intensity or recording conditions did not skew the feature representations.

For video data, pixel intensity values were normalized to fall within a range of 0 to 1 by dividing by 255 (the maximum pixel value in an 8-bit RGB image). This normalization step ensured uniformity in video inputs, which is critical for deep learning models like convolutional neural networks (CNNs). Additionally, preprocessing included resizing frames to a fixed resolution (e.g., 224×224) to standardize input dimensions across all video clips.

Each participant's data was split into training and testing subsets, with 70% of the trials allocated for training and 30% for testing. For EEG, this resulted in 140 trials for training and 60 for testing per participant. Each trial, segmented into 20-second intervals, was further divided into smaller temporal slices (e.g., 5 seconds) for specific tasks, increasing the number of training instances. Similarly, audio data was split, with 280 audio segments (5-second intervals) used for training

and 120 for testing. Video data followed the same proportion, ensuring a balanced representation of all emotion classes across the training and testing sets. The splits were designed to preserve temporal consistency, preventing overlap between training and testing segments within the same trial to avoid data leakage.

B. Proposed Architecture and Methodology

Figure 1 illustrates the proposed network architecture (MFAV).

Audio: The audio modality processing begins with the transformation of raw audio features into high-dimensional token representations, enriched with positional context to extract the temporal structure of the input. Raw audio features, such as MFCCs, are represented as x_i , where i indexes the time step, and d_x is the dimensionality of the feature vector. Each feature vector is mapped to a higher-dimensional token space using a learnable embedding function $f_{embed}: R^{d_x} \rightarrow R^{d_t}$. Positional encodings are added to the tokens to retain temporal information. The positional encodings are defined as:

$$PE_{(i,2k)} = \sin\left(\frac{i}{10000^{2k/d_t}}\right), PE_{(i,2k+1)} = \cos\left(\frac{i}{10000^{2k/d_t}}\right) \quad (1)$$

where i is the time index, k is the feature index, and d_t is the embedding dimensionality. The resulting token at each time step is then computed as $T_i = f_{embed}(x_i) + PE_i$. This operation produces a token matrix $T \in R^{n \times d_t}$, where n is the number of audio frames. The addition of positional encoding ensures that the temporal sequence of audio features is preserved and incorporated into the model's processing pipeline. Once the tokens are generated, spatial channel attention is applied to dynamically re-weight each token's importance of different feature dimensions. This mechanism identifies which features are most critical for the task, such as pitch, tone, or intensity variations, particularly relevant for emotion recognition. The attention mechanism computes weights for each channel (or feature dimension) using $A_{spatial} = \text{softmax}(W_s \cdot T^T + b_s)$,

where $W_s \in R^{d_t \times d_t}$ is a learnable weight matrix, $T^T \in R^{d_t \times n}$ is the transposed token matrix, and $b_s \in R^{d_t}$ is a bias vector. These attention weights are then applied to the token matrix $T_{attended} = A_{spatial}^T \odot T$

where $\odot$ denotes element-wise multiplication. This process amplifies the most relevant features and suppresses irrelevant ones, producing an attention-weighted token matrix $T_{attended} \in R^{n \times d_t}$. This step ensures that the most significant aspects of the audio input, such as spectral patterns corresponding to specific emotions, are emphasized for further processing. To condense the sequence of tokens into a compact representation, the model employs a mixer that aggregates the attention-weighted tokens across the temporal dimension. The mixer computes a summary vector using a learnable weighted aggregate.

$$M = W_m \cdot \sum_{i=1}^n T_{attended}[i, :] + b_m, \quad (2)$$

where $W_m \in R^{d_t \times d_m}$ is a learnable projection matrix, $b_m \in R^{d_m}$ and $M \in R^{d_m}$ is the resulting compact representation. This operation reduces the temporal dimension by summarizing the sequence into a single vector while retaining the critical features identified during the attention step. The mixer ensures that the audio features are compactly

represented, facilitating efficient integration with downstream components. The compact representation M is then employed to produce query (Q), key (K), and value (V) matrices, which serve as inputs to the attention mechanism in subsequent stages. These matrices are computed as $Q = W_q \cdot M$, $K = W_k \cdot M$, $V = W_v \cdot M$,

where $W_q, W_k, W_v \in R^{d_m \times d_a}$ are learnable projection matrices, and d_a is the dimensionality of the attention mechanism. The Q matrix represents the features being queried, K represents the features to compare against, and V carries the feature values. These representations enable the model to compute relationships between audio features and other modalities in later stages. Finally, the feature extraction block refines the aggregated audio representation into a format suitable for cross-modal fusion. This step involves projecting the compact representation M into a fusion-compatible feature space and normalizing the resulting vector. The projection is performed using $F_{proj} = \text{Norm}(W_{proj} \cdot M + b_{proj})$ where $W_{proj} \in R^{d_m \times d_f}$ and $b_{proj} \in R^{d_f}$ are learnable parameters. The output $F_{proj} \in R^{d_f}$ is a refined, normalized vector that serves as the audio modality's input to the fusion layer.

Video: The video modality processing begins with extracting frame-wise local embeddings, which transform raw video frames into high-dimensional token representations. Each video frame is represented as a spatial matrix $v_i \in$

$R^{H \times W \times C}$. These frames are flattened into vectors and passed through a learnable embedding function $g_{embed}: R^{H \times W \times C} \rightarrow R^{d_t}$, mapping each frame into a token of dimension d_t . The token representation of a frame is given by: $T_i = g_{embed}(v_i)$ where $T_i \in R^{d_t}$ represents the token corresponding to the i^{th} frame. The output of this stage is a sequence of tokens $T \in R^{n \times d_t}$ where n is the number of frames in the video. These tokens capture spatial features within individual frames, providing a local representation of the video data.

The next step applies multi-head self-attention to extract global relationships within and between frames in the video. This enables the network to identify dependencies across the entire sequence of frames by computing attention scores between all token pairs. The query (Q), key (K), and value (V) matrices are generated from the tokens $Q = W_q \cdot T$, $K = W_k \cdot T$, $V = W_v \cdot T$. The attention values are calculated as $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_a}}\right)V$ where softmax ensures that the attention weights sum to 1. Multi-head attention splits the input into multiple heads, allowing the model to simultaneously focus on different aspects of the frame sequence. The outputs of all heads are combined and projected back to the original token dimension d_t . This process generates a refined token matrix, where global temporal relationships between frames are encoded. Next, the feed-forward network refines the spatial and temporal features learned in the attention mechanism. The refined tokens are computed as $T_{refined} = \text{ReLU}(W_{ff} \cdot T_{attended} + b_{ff})$.

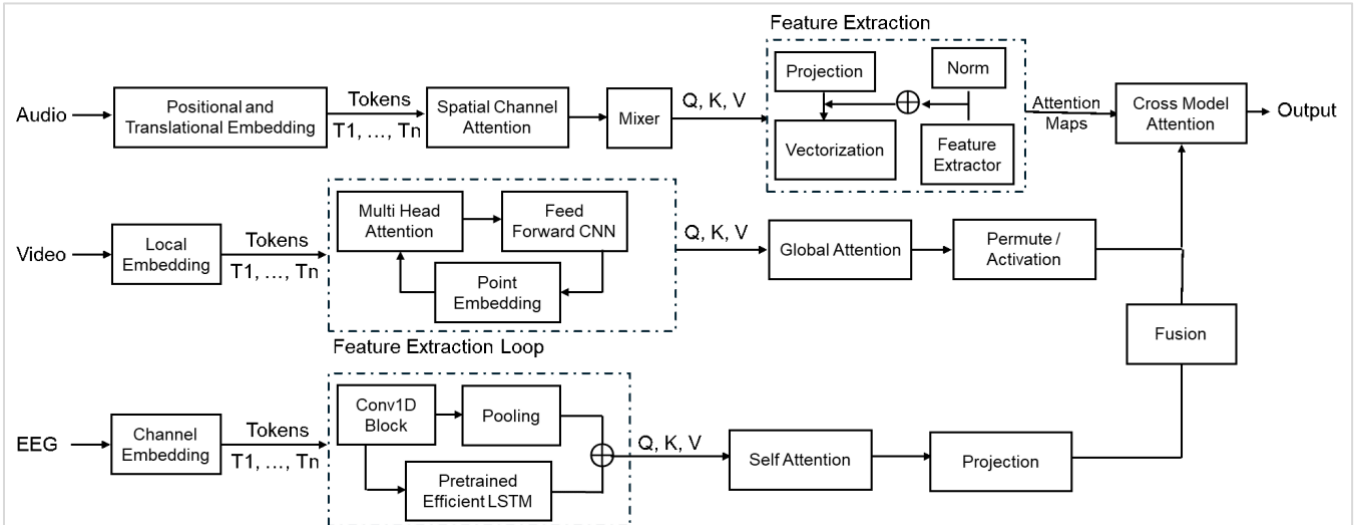


Fig. 1. Proposed Architecture of Multifold Fusion Attention Variant (MFAV)

This operation ensures that each token is enriched with non-linear transformations, making the feature representations more expressive and suitable for downstream tasks. The refined tokens maintain the same shape. A point embedding mechanism is then used to enhance positional information and ensure that the sequence order of frames is retained. Point embeddings are similar to positional embeddings in the audio modality, adding trainable embeddings to each token based on its position in the sequence $T_{final}[i] = T_{refined}[i] + p_i$ where $p_i \in R^{d_t}$ is the positional embedding for the i^{th} frame. This step ensures that the temporal order of frames is incorporated into the model's processing. The final sequence of tokens is passed through a global attention mechanism that

computes a single representation of the video by aggregating information from all frames. Finally, the compact representation M is passed through a feature extraction block to prepare it for cross-modal fusion. This involves a linear projection and normalization. The output $F_{proj} \in R^{d_f}$ is a refined feature vector that encodes both spatial and temporal information from the video modality, making it compatible for integration with other modalities.

EEG: The EEG modality processing begins by transforming raw EEG signals into token representations. Each EEG trial consists of signals recorded from multiple channels over time, represented as $e_{i,c} \in R^T$, where T is the

number of time points, i is the time index, and c is the channel index. A learnable channel embedding function $R^T \rightarrow R^{d_t}$ maps the EEG signals into a high-dimensional token space $T_{i,c} = h_{embed}(e_{i,c})$, producing tokens $T \in R^{n \times C \times d_t}$, where n is the number of time segments, C is the number of channels. This step preserves the spatiotemporal characteristics of the EEG data across all channels. A 1D convolutional block is applied to the tokenized EEG signals to capture local temporal patterns. The convolution operation is defined as: $F_{conv} = \text{Conv1D}(T)$ where Conv1D operates along the temporal axis. This produces feature maps that highlight temporal dependencies within each channel. The output is pooled across the temporal dimension using average pooling to reduce the sequence length while retaining critical temporal information $F_{pooled} = \text{Pool}(F_{conv})$. The pooled features are further processed by a trained Efficient LSTM network, which models long-term temporal dependencies in the EEG signals. The LSTM computes $h_t, c_t = \text{LSTM}(F_{pooled}[t], h_{t-1}, c_{t-1})$ where h_t and c_t are the hidden and cell states at time t . The final hidden state h_T represents the condensed spatiotemporal features of the EEG data. Self-attention is applied to the output of the LSTM to capture inter-channel dependencies. The query (Q), key (K), and value (V) matrices are then computed. The self-attention mechanism computes the attended. This output captures channel-wise relationships and interactions, producing a globally attended representation.

Finally, the attended EEG representation is passed through a projection layer and normalization to prepare it for cross-modal fusion: $F_{proj} = \text{Norm}(W_{proj} \cdot F_{attended} + b_{proj})$.

Fusion: In transformer fusion, modality-specific features are concatenated into a unified sequence and processed through self-attention, capturing global and sequential cross-modal interactions. Attention-based fusion computes cross-modal dependencies by treating each modality's features as queries, keys, and values, emphasizing the most relevant interactions, such as aligning audio pitch with facial expressions or EEG patterns. CNN fusion treats modality features as separate channels, aggregating spatial and localized correlations through convolutional layers and effectively capturing hierarchical relationships.

The final fusion combines modality features $F_{audio}, F_{video}, F_{eeg}$ by concatenation $F_{fusion} = \text{Concat}(F_{audio}, F_{video}, F_{eeg})$, followed by projection: $F_{unified} = W_{fusion} \cdot F_{fusion} + b_{fusion}$.

C. Experiments and Evaluation

We compared the proposed model (MFAV) with four other popular models - TransNet, CNNNet, MULT and MultiTrans [20][21][22][23]. All the models were trained on the same training and testing dataset using Adam's optimizer for 50 epochs, with a batch size of 32, a learning rate of 0.04, a momentum of 0.9, and a weight decay of $1e-3$. The models were evaluated using accuracy (no. of correctly predicted classes) and mean absolute error (MAE), which helps evaluate both classification correctness and prediction consistency, making it suitable for multimodal emotion recognition tasks.

IV. RESULTS AND DISCUSSION

The performance metrics for all the models are summarized in Table I.

TABLE I. PERFORMANCE METRICS

Model	Accuracy (%)	MAE
MFAV	91.2	0.08
TransNet	85.3	0.15
CNNNet	82.7	0.18
MULT	84.1	0.16
MultiTrans	83.5	0.17

It is seen from Table I that MFAV demonstrates an average accuracy improvement of 7.39% across all models with a significant enhancement ($p < 0.05$). This demonstrates that while conventional models perform reasonably well, they are less effective in capturing the complexity of multimodal data compared to MFAV. MFAV learns directly from raw data and dynamically focuses on relevant features across modalities. This enables the MFAV to integrate complementary information from multimodal inputs effectively, resulting in higher accuracy and reduced error. Table II summarizes the effect of different fusion mechanisms on the MFAV model.

TABLE II. SIMULATION IMPACT OF FUSION TECHNIQUES ON MFAV

Model	Accuracy (%)
Summation	85.5
CNN	86.7
Transformer	88.3
Attention	90.1

As shown in Table II, there is a clear progression in performance from summation to the proposed attention-based fusion. Summation, while straightforward, results in the lowest accuracy and the highest number of false positives. This can be attributed to its simplistic approach of directly concatenating input features, which limits the model's ability to capture complex inter-modal interactions. CNN fusion performs slightly better. This improvement comes from processing modalities independently and combining their outputs at the decision level. However, this technique fails to effectively leverage the shared representations between modalities, as it treats them in isolation until the final stages of processing. Transformer fusion demonstrates a further improvement and a significant reduction in false positives. Integrating features at mid-layers allows the model to learn representations across modalities, capturing richer inter-modal dependencies jointly. However, it lacks the dynamic adaptability to attend to modality-specific nuances. The attention-based intermediate fusion proposed in this work outperforms all other techniques, achieving the highest accuracy (90.1%) and the lowest false positives. This approach dynamically assigns attention weights to modality-independent features, enabling the network to focus on the most significant information from each modality. Table III summarizes the ablation study on MFAV.

The ablation study highlights the critical role of various components in the model's performance. The baseline model, which integrates all features, achieves the highest accuracy. Removing key components such as positional/translational embeddings, spatial channel attention, or cross-model attention leads to significant performance drops, demonstrating their importance in capturing spatial, temporal, and inter-modal relationships. The pretrained LSTM contributes notably to sequential feature learning, as its removal reduces accuracy. Similarly, the absence of intermediate fusion results in the lowest accuracy, underscoring the importance of effective feature integration.

These findings emphasize the synergistic impact of all components in achieving optimal model performance. Fig. 2 shows the uncertainty graph over training epochs for all the models

TABLE III. RESULTS OF ABLATION STUDY ON MFAV

Model	Accuracy (%)
Full Model (Baseline)	91.2
No Positional/Translational Embedding	89.3
No Spatial-Channel Attention	88.4
No Cross Model Attention	87.1
No Global Attention	87.5
No LSTM	87.3
No Multihead Attention	87.7
Only Local Embedding	87.0
No Fusion	83.2

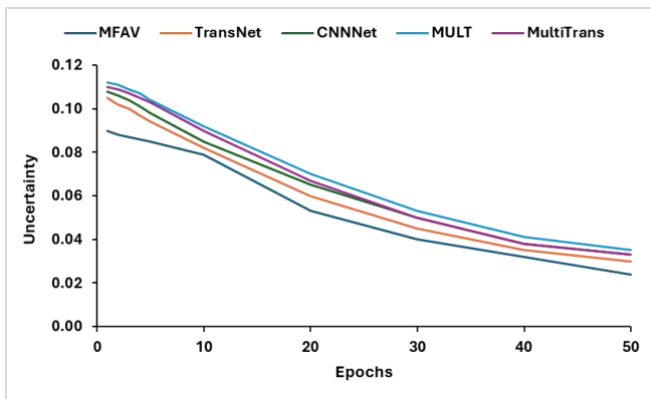


Fig. 2. Uncertainty during Training

The proposed model, MFAV, demonstrates the lowest uncertainty consistently throughout the training process, indicating its robustness and reliability. While all models show a decreasing trend in uncertainty as training progresses, MFAV achieves the steepest decline, stabilizing at much lower values compared to the other models.

TransNet and CNNNet exhibit slightly higher uncertainty than MFAV but remain relatively competitive. MULT and MultiTrans, on the other hand, show the highest uncertainty levels throughout, reflecting their less robust handling of the data variability. The gap between MFAV and other models becomes more pronounced in later epochs, highlighting the advantage of the attention-based fusion mechanism and end-to-end learning employed in MFAV.

This consistent performance improvement in MFAV underscores its superior capability to minimize uncertainty and better generalize across the data, making it a highly effective solution for multimodal tasks in real-world applications. An additional overview of the model's hyper-parameters is tabulated in Table IV.

TABLE IV. RESULTS OF ABLATION STUDY ON MFAV

Data	Value	Hyper-parameter	Accuracy (%)
Train Size	70% (140 EEG trials, 280 audio segments, 200 video sequences)	Learning Rate: 0.04, Batch: 32, Epochs: 50	98.8
Test Size	30% (60 EEG trials, 120 audio segments, 100 video sequences)	Momentum: 0.9, Weight Decay: 1e-3	91.2

V. CONCLUSIONS

Thus, we proposed multifold attention multi-modal learning for enhanced emotion recognition across various modalities. This method improves the accuracy of recognizing emotions and lays a robust foundation for incorporating emotional understanding into foundational or generative models. This study's limitations include the dataset's restricted diversity, which may not fully capture the variability in real-world emotional expressions. Additionally, the model's generalizability remains unconfirmed across different datasets, posing a risk of overfitting to the specific characteristics of the EAV dataset. Future work will focus on refining emotion-aware mechanisms in generative models to better detect and respond to emotional cues. Key goals include expanding emotional range, improving context retention, and enabling personalized interactions. We will also address ethical concerns, ensuring fair and empathetic AI for applications like mental health and customer support.

REFERENCES

- [1] Houssein, E.H., Hammad, A. & Ali, A.A. "Human emotion recognition from EEG-based brain-computer interface using machine learning: a comprehensive review." *Neural Comput & Applic* 34, 12527–12557 (2022)..
- [2] S. Kalateh, L. A. Estrada-Jimenez, S. Nikghadam-Hojjati and J. Barata, "A Systematic Review on Multimodal Emotion Recognition: Building Blocks, Current State, Applications, and Challenges," in *IEEE Access*, vol. 12, pp. 103976-104019, 2024.
- [3] Kosti, R., Alvarez, J. M., Recasens, A., & Lapedriza, A. (2020). "Context-Based Emotion Recognition using EMOTIC Dataset."
- [4] Sharma, R., Dharmyal, H., Raj, B., & Singh, R. (2022). "Unifying the Discrete and Continuous Emotion labels for Speech Emotion Recognition."
- [5] Machová, K., Szabóová, M., Paralič, J., & Mičko, J. (2023). "Detection of emotion by text analysis using machine learning. *Frontiers in Psychology*, 14, 1190326."
- [6] Lee, J., Kim, J., & Kim, H. (2024). "Emotion Recognition Using EEG Signals and Audiovisual Features with Contrastive Learning." *Bioengineering*, 11(10), 997.
- [7] Salas-Cáceres, J., Lorenzo-Navarro, J., Freire-Obrégón, D. et al. "Multimodal emotion recognition based on a fusion of audiovisual information with temporal dynamics. *Multimed Tools Appl* (2024)."
- [8] Li, D., Wang, Y., Funakoshi, K., & Okumura, M. (2023). "Joyful: Joint Modality Fusion and Graph Contrastive Learning for Multimodal Emotion Recognition."
- [9] Palash, M., & Bhargava, B. (2023). "EMERSK -- Explainable Multimodal Emotion Recognition with Situational Knowledge."
- [10] Cheng, Z., Cheng, Z., He, J., Sun, J., Wang, K., Lin, Y., Lian, Z., Peng, X., & Hauptmann, A. (2024). "Emotion-LLaMA: Multimodal Emotion Recognition and Reasoning with Instruction Tuning."
- [11] R, J. D., Venkatraman, S., Narendran, M., Sharma, V., Malarvannan, S., & Gandomi, A. H. (2024). "Multimodal Emotion Recognition using Audio-Video Transformer Fusion with Cross Attention."
- [12] Pan, B., Hirota, K., Jia, Z. et al. "Multimodal emotion recognition based on feature selection and extreme learning machine in video clips." *J Ambient Intell Human Comput* 14, 1903–1917 (2023).
- [13] Jia, N., Zheng, C. & Sun, W. "A multimodal emotion recognition model integrating speech, video and MoCAP." *Multimed Tools Appl* 81, 32265–32286 (2022).

- [14] S. Siriwardhana, T. Kaluarachchi, M. Billinghamurst and S. Nanayakkara, "Multimodal Emotion Recognition With Transformer-Based Self Supervised Feature Fusion," in *IEEE Access*, vol. 8, pp. 176274-176285, 2020.
- [15] H. -D. Le, G. -S. Lee, S. -H. Kim, S. Kim and H. -J. Yang, "Multi-Label Multimodal Emotion Recognition With Transformer-Based Fusion and Emotion-Level Representation Learning," in *IEEE Access*, vol. 11, pp. 14742-14751, 2023.
- [16] Shayaninasab, M., & Babaali, B. (2024). "Multi-Modal Emotion Recognition by Text, Speech and Video Using Pretrained Transformers."
- [17] Li, L., Chen, Jh. (2006). "Emotion Recognition Using Physiological Signals." In: Pan, Z., Cheok, A., Haller, M., Lau, R.W.H., Saito, H., Liang, R. (eds) *Advances in Artificial Reality and Tele-Existence*. ICAT 2006.
- [18] Nagrani, A., Yang, S., Arnab, A., Jansen, A., Schmid, C., & Sun, C. (2021). "Attention Bottlenecks for Multimodal Fusion."
- [19] Lee, M., Shomanov, A., YAZICI, A., Kabidenova, Z., Begim, B., Nyssanbay, A., & Lee, S.-W. (2023). "EAV: EEG-Audio-Video Dataset for Emotion Recognition in Conversational Contexts (0.1.0) [Data set]. Zenodo."
- [20] J. Huang, J. Tao, B. Liu, Z. Lian, and M. Niu, "Multimodal transformer fusion for continuous emotion recognition," in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 3507–3511.
- [21] Begazo R, Aguilera A, Dongo I, Cardinale Y. A Combined CNN Architecture for Speech Emotion Recognition. *Sensors (Basel)*. 2024 Sep 6;24(17):5797. doi: 10.3390/s24175797. PMID: 39275707; PMCID: PMC11398044.
- [22] Y.-H. H. Tsai, S. Bai, P. P. Liang, J. Z. Kolter, L.-P. Morency, and R. Salakhutdinov, "Multimodal transformer for unaligned multimodal language sequences," in *Proceedings of the conference. Association for Computational Linguistics. Meeting*, vol. 2019. NIH Public Access, 2019, p. 6558.
- [23] Miah MSU, Kabir MM, Sarwar TB, Safran M, Alfarhood S, Mridha MF. A multimodal approach to cross-lingual sentiment analysis with ensemble of transformer and LLM. *Sci Rep*. 2024 Apr 26;14(1):9603. doi: 10.1038/s41598-024-60210-7. PMID: 38671064; PMCID: PMC11053029.